Archana Thanuwana

SOFTWARE ENGINEERING & IT UNDERGRADUATE

AI/ML Enthusiast • Full-Stack Developer • Systems Builder

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PROFESSIONAL SUMMARY

Information Technology undergraduate at the University of Moratuwa with solid expertise in software engineering, autonomous AI agent architectures, and full-stack development. Experienced in building multi-agent AI workflows (LangGraph, LangChain), RAG pipelines, FastAPI backends, and modular C/C++ systems. Passionate about writing clean, scalable code and delivering innovative software solutions.

EDUCATION

University of Moratuwa, Sri Lanka

2024 – Present

B.Sc. (Hons) in Information Technology — Faculty of Information Technology

Core Coursework: Programming Fundamentals (C/C++), Data Structures & Algorithms, Object-Oriented Programming (Java), Database Management Systems (SQL), Software Engineering, Artificial Intelligence.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, Dart, Go, JavaScript, TypeScript, SQL

Frameworks: FastAPI, Streamlit, Flutter, Angular, SQLAlchemy

AI & LLMs: LangChain, LangGraph, Adaptive RAG, Agentic Workflows, MCP

Databases & Tools: PostgreSQL, SQLite, Git, GitHub, Linux, VS Code, Postman

SELECTED PROJECTS

Personal AI Assistant — Multi-Agent Architecture

Python, LangGraph, LangChain

- Engineered an autonomous multi-agent assistant utilizing LangGraph state machines and LangChain tooling.
- Implemented dynamic intent routing to dedicated news, scam-detection, and general-purpose agents with Tavily search and Gradio UI.

LangGraph | LangChain | Tavily | Gradio

Database Copilot Agent

FastAPI, PostgreSQL, SQLAlchemy

- Built an intelligent database assistant enabling natural language interaction with PostgreSQL via automated SQL query generation and schema validation.

FastAPI | PostgreSQL | SQLAlchemy | LLM Agents

Adaptive RAG Architecture

Python, Vector Search, LLM

- Engineered an Adaptive Retrieval-Augmented Generation pipeline using query evaluation, self-correction, and dynamic retrieval routing.

Adaptive RAG | Vector Search | Python

WhatsApp MCP Server

Go (Golang), MCP Protocol

- Developed an open Model Context Protocol (MCP) server in Go exposing WhatsApp messaging tools for AI agent applications.

Go | MCP | APIs

Disaster Relief & Aid Distribution System

C, Data Structures, File I/O

- Engineered a modular console application modeling Sri Lanka DMC emergency operations with 5 integrated subsystems and binary file persistence.

C Programming | Binary File I/O | Algorithms

Money Management Mobile Application

Flutter, Dart

- Created a cross-platform personal finance mobile app for real-time expense tracking, budget allocation, and categorization.

Flutter | Dart | Mobile UX

EXTRACURRICULAR & INTERESTS

- **Open-Source & Hackathons:** Active open-source builder; exploring IoT systems (smart shoe dryer controller, automated dispenser).
- **Languages:** English (Professional Working Proficiency), Sinhala (Native).